Subject: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2021 – Paper 3 – Solutions**

June 2021 – Question 1 – Solution

(a) The roots of the cubic equation $2x^3 - 3x^2 - x + 8 = 0$ are α, β and γ. Determine the equation which has roots α^2, β^2 and γ^2.

Note: $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$(\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$ [7]

$$2x^3 - 3x^2 - x + 8 = 0$$

$$(\div 2) \qquad x^3 - \frac{3}{2}x^2 - \frac{1}{2}x + 4 = 0$$

$$\alpha + \beta + \gamma = \frac{3}{2}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{1}{2}$$

$$\alpha\beta\gamma = -4$$

For the new roots:

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= \left(\frac{3}{2}\right)^2 - 2\left(-\frac{1}{2}\right)$$

$$= \frac{9}{4} + 1$$

$$= \frac{13}{4}$$

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= \left(-\frac{1}{2}\right)^2 - 2(-4)\left(\frac{3}{2}\right)$$

$$= \frac{1}{4} + 12$$

$$= \frac{49}{4}$$

$$\alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2$$

$$= (-4)^2$$

$$= 16$$

Hence, the new equation whose roots are α^2, β^2 and γ^2 is:

$$x^3 - (sum\ of\ roots)x^2 + (sum\ of\ the\ product\ of\ two\ of\ the\ roots\ at\ a\ time)x - (product\ of\ the\ roots) = 0$$

$$x^3 - \frac{13}{4}x^2 + \frac{49}{4}x - 16 = 0$$

$(\times 4)$

$$4x^3 - 13x^2 + 49x - 64 = 0$$

(b) Prove by mathematical induction that $1 + 3 + 3^2 + 3^3 + \cdots + 3^{n-1} = \frac{3^n - 1}{2}$ for all positive integers $n \geq 1$. [8]

Let $P(n)$ be the statement $1 + 3 + 3^2 + 3^3 + \cdots + 3^{n-1} = \frac{3^n - 1}{2}, \forall n \in \mathbb{Z}^+, n \geq 1$

Looking at $P(1)$:

L.H.S $= 1$

R.H.S. $= \frac{3^{(1)}-1}{2}$

$= \frac{2}{2}$

$= 1$

L.H.S. = R.H.S.

$\therefore P(n)$ is true for $n = 1$.

Assume $P(k)$ is true.

That is, $1 + 3 + 3^2 + 3^3 + \cdots + 3^{k-1} = \frac{3^k-1}{2}$

Looking at $P(k+1)$.

That is, $1 + 3 + 3^2 + 3^3 + \cdots + 3^k = \frac{3^{k+1}-1}{2}$

We need to prove $P(k+1)$.

Taking L.H.S:

$1 + 3 + 3^2 + 3^3 + \cdots + 3^k = [1 + 3 + 3^2 + 3^3 + \cdots + 3^{k-1}] + 3^k$

$= \frac{3^k-1}{2} + 3^k$

$= \frac{1}{2}(3^k - 1 + 2 \cdot 3^k)$

$= \frac{1}{2}(3 \cdot 3^k - 1)$

$= \frac{1}{2}(3^{k+1} - 1)$

$= \frac{3^{k+1}-1}{2}$

= R.H.S.

$\therefore$ If $P(k)$ is true, then $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$1 + 3 + 3^2 + 3^3 + \cdots + 3^{n-1} = \frac{3^n - 1}{2}, \forall n \in \mathbb{Z}^+, n \geq 1$

Q.E.D.

(c) Solve the equation $\log_3(5x-2) + \log_9(5x-2) = 3$, where $x > \frac{2}{5}$. [5]

$$\log_3(5x-2) + \log_9(5x-2) = 3$$

$$\log_3(5x-2) + \frac{\log_3(5x-2)}{\log_3 9} = 3$$

$$\log_3(5x-2) + \frac{\log_3(5x-2)}{2} = 3$$

$$2\log_3(5x-2) + \log_3(5x-2) = 6$$

$$\log_3(5x-2)^2 + \log_3(5x-2) = 6$$

$$\log_3(5x-2)^2(5x-2) = 6$$

$$\log_3(5x-2)^3 = 6$$

Converting to exponential form gives:

$$(5x-2)^3 = 3^6$$

$$[(5x-2)^3]^{\frac{1}{3}} = [3^6]^{\frac{1}{3}}$$

$$5x - 2 = 3^2$$

$$5x - 2 = 9$$

$$5x = 9 + 2$$

$$5x = 11$$

$$x = \frac{11}{5}$$

Total: 20 marks

June 2021 – Question 2 – Solution

(a) $P(1,5,2)$, $Q(2,6,3)$ and $R(4,2,-1)$ are the vertices of a triangle.

(i) Determine the displacement vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$. [4]

The points are $P(1,5,2)$, $Q(2,6,3)$ and $R(4,2,-1)$.

The position vectors are $\overrightarrow{OP} = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$, $\overrightarrow{OQ} = \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix}$ and $\overrightarrow{OR} = \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$.

Using the triangle law,

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

$$= \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Using the triangle law,

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP}$$

$$= \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ -3 \\ -3 \end{pmatrix}$$

$\therefore$ The displacement vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are $\overrightarrow{PQ} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\overrightarrow{PR} = \begin{pmatrix} 3 \\ -3 \\ -3 \end{pmatrix}$.

(ii) Hence, determine $|\overrightarrow{PQ}|$ and $|\overrightarrow{PR}|$. [4]

$$|\overrightarrow{PQ}| = \sqrt{(1)^2 + (1)^2 + (1)^2}$$

$$= \sqrt{1 + 1 + 1}$$

$$= \sqrt{3} \text{ units}$$

$$|\overrightarrow{PR}| = \sqrt{(3)^2 + (-3)^2 + (-3)^2}$$

$$= \sqrt{9 + 9 + 9}$$

$$= \sqrt{9 \times 3}$$

$$= 3\sqrt{3} \text{ units}$$

(iii) Determine the acute angle between $\overrightarrow{PQ}$ and $\overrightarrow{PR}$. [4]

$$\overrightarrow{PQ} \cdot \overrightarrow{PR} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ -3 \end{pmatrix}$$

$$= (1)(3) + (1)(-3) + (1)(-3)$$

$$= -3$$

Now,

$$\cos\theta = \frac{\overrightarrow{PQ} \cdot \overrightarrow{PR}}{|\overrightarrow{PQ}||\overrightarrow{PR}|}$$

$$\cos\theta = \frac{-3}{(\sqrt{3})(3\sqrt{3})}$$

$$\cos\theta = -\frac{1}{3}$$

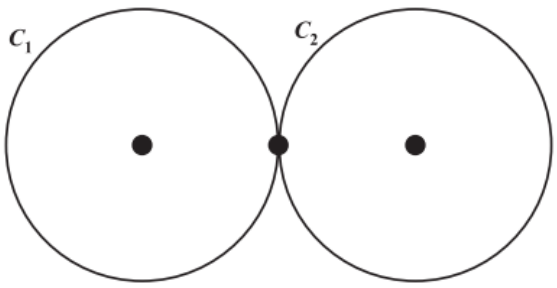
$$\theta = \cos^{-1}\left(-\frac{1}{3}\right)$$

$$\theta = 109.5°$$

(b) The following diagram (**not drawn to scale**) shows two circles, C_1 and C_2, touching at the point $P(-4, 3)$.

The equation of the circle C_1 is given by $x^2 + y^2 + 14x - 6y + 49 = 0$.

The radius of C_2 is 3. Determine the equation of the circle C_2. [8]



The equation of C_1 is given by:

$$x^2 + y^2 + 14x - 6y + 49 = 0$$

$$x^2 + 14x + y^2 - 6y = -49$$

$$x^2 + 14x + 49 + y^2 - 6y + 9 = -49 + 49 + 9$$

$$(x + 7)^2 + (y - 3)^2 = 9$$

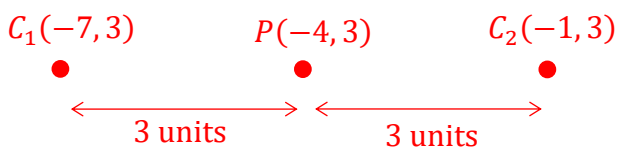
which is in the form $(x - a)^2 + (y - b)^2 = r^2$

where the centre is $C(-7, 3)$ and radius, $r = 3$.

The radius of both circles is 3 units.

The point $P(-4, 3)$ is the midpoint of the centre of the circles.

Consider the diagram below:



The centre of C_2 is $(-1, 3)$.

The radius of C_2 is 3 units.

Hence, the standard form of the equation of C_2 is:

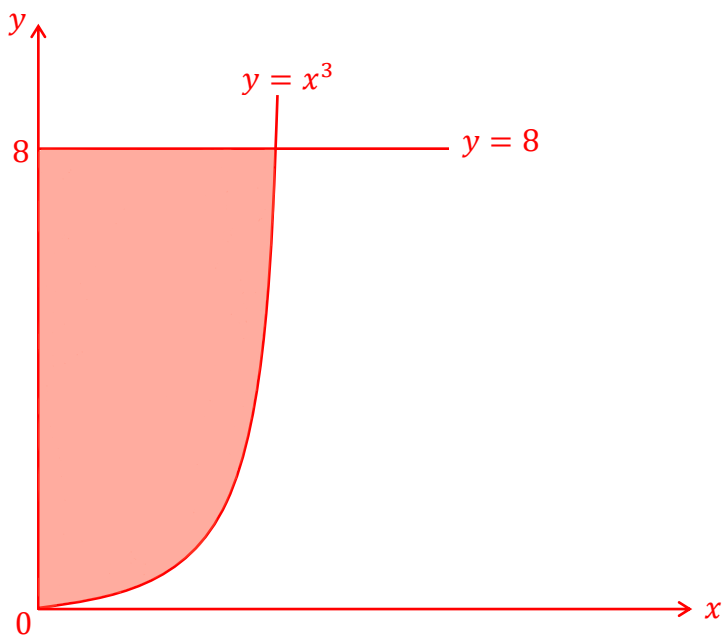
$(x + 1)^2 + (y - 3)^2 = 9$

Total: 20 marks

June 2021 – Question 3 – Solution

(a) Calculate the volume of the solid obtained by rotating the region bounded by the curves $y = x^3, y = 8$ and $x = 0$ around the y-axis. [6]

Consider the sketch below:



Volume around the y-axis $= \pi \int_a^b x^2 \, dy$

Now,

$$y = x^3$$

$$x^3 = y$$

$$(x^3)^{\frac{2}{3}} = y^{\frac{2}{3}}$$

$$x^2 = y^{\frac{2}{3}}$$

$$V = \pi \int_a^b x^2\, dy$$

$$= \pi \int_0^8 y^{\frac{2}{3}}\, dy$$

$$= \pi \left[\frac{y^{\frac{5}{3}}}{\left(\frac{5}{3}\right)}\right]_0^8$$

$$= \frac{3\pi}{5}\left[y^{\frac{5}{3}}\right]_0^8$$

$$= \frac{3\pi}{5}\left[(8)^{\frac{5}{3}} - (0)^{\frac{5}{3}}\right]$$

$$= \frac{3\pi}{5}(32 - 0)$$

$$= \frac{96\pi}{5} \text{ units}^3$$

(b) The voltage, V, in a circuit can be represented by the equation $\frac{dV}{dt} - \frac{V}{2} = 0$.

Given that $V = 25$ volts and $t = 0$, determine an expression for V in terms of t.

Note that $\int \frac{1}{x}\, dx = \ln x + c$. [6]

$$\frac{dV}{dt} - \frac{V}{2} = 0$$

$$\frac{dV}{dt} = \frac{V}{2}$$

$$\frac{dV}{dt} = \frac{\left(\frac{1}{2}\right)}{\left(\frac{1}{V}\right)}$$

Separating variables:

$$\frac{1}{V}\, dV = \frac{1}{2}\, dt$$

Integrating both sides gives:

$\int \frac{1}{V}\, dV = \int \frac{1}{2}\, dt$

$\ln V = \frac{1}{2}t + c$ (G.S.)

When $V = 25$ and $t = 0$,

$\ln 25 = \frac{1}{2}(0) + c$

$x = \ln 25$

So, we have,

$\ln V = \frac{1}{2}t + \ln 25$ (P.S.)

$e^{\ln V} = e^{\frac{1}{2}t + \ln 25}$

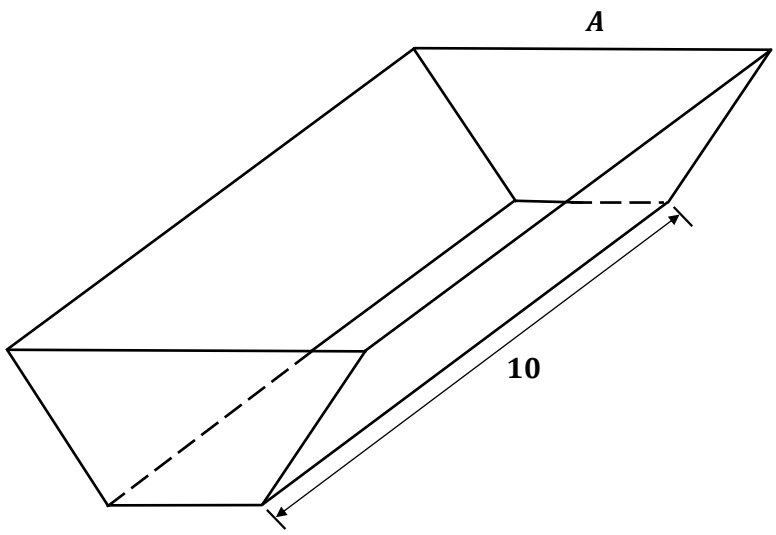
$V = e^{\frac{1}{2}t} \cdot e^{\ln 25}$

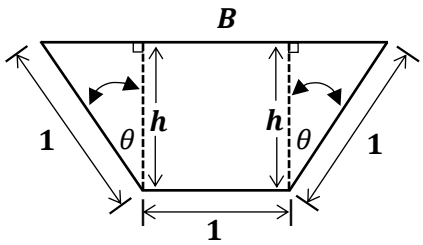
$V = e^{\frac{1}{2}t} \cdot 25$

$V = 25e^{\frac{t}{2}}$

$\therefore$ An expression of V in terms of t is: $V = 25e^{\frac{t}{2}}$

(c) Diagram A below (**not drawn to scale**) shows the design of a trough. The cross-section of the trough, which has the shape of a trapezium, is shown in Diagram B. All lengths are in metres.

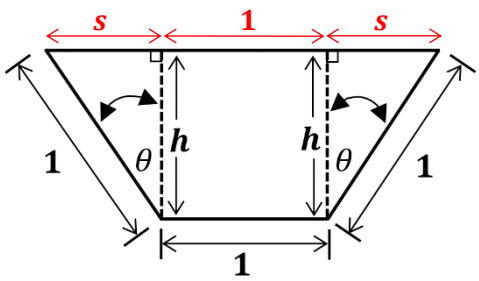




The trough must be made using the dimensions shown, but the angle θ may vary.

(i) Show that the cross-sectional area, X, of the trough is

$$X = \cos\theta\,(1 + \sin\theta)$$ [3]



Now,

$\cos\theta = \frac{adj}{hyp}$

$\cos\theta = \frac{h}{1}$

$h = \cos\theta$

And,

$\sin\theta = \frac{opp}{hyp}$

$\sin\theta = \frac{s}{1}$

$s = \sin\theta$

Area of trapezium $= \frac{1}{2}(a+b)h$

$= \frac{1}{2}(s+1+s+1)h$

$= \frac{1}{2}(2s+2)h$

$= \frac{1}{2}(2\sin\theta+2)\cos\theta$

$= \frac{1}{2}(2)(\sin\theta+1)\cos\theta$

$= (\sin\theta+1)\cos\theta$

$= \cos\theta(1+\sin\theta)$

$\therefore$ The cross-sectional area of the trough, $X = \cos\theta(1+\sin\theta)$.

Q.E.D.

(ii) Hence, or otherwise, determine the acute angle, θ, that would **maximize** the volume of the trough. [5]

Length of the trough = 10 metres

Hence,

Volume of the trough, V = Cross-sectional Area × length of trough

$$= \cos\theta(1 + \sin\theta) \times 10$$

$$= 10\cos\theta\,(1 + \sin\theta)$$

$V = 10\cos\theta\,(1 + \sin\theta)$

Let $u = 10\cos\theta$, $v = 1 + \sin\theta$

$\frac{du}{d\theta} = -10\sin\theta$ $\frac{dv}{d\theta} = \cos\theta$

$$\frac{dV}{d\theta} = u\frac{dv}{d\theta} + v\frac{du}{d\theta}$$

$$= 10\cos\theta\,(\cos\theta) + (1 + \sin\theta)(-10\sin\theta)$$

$$= 10\cos^2\theta - 10\sin\theta - 10\sin^2\theta$$

At stationary points, $\frac{dV}{d\theta} = 0$.

When $\frac{dV}{d\theta} = 0$,

$$10\cos^2\theta - 10\sin\theta - 10\sin^2\theta = 0$$

$$\cos^2\theta - \sin\theta - \sin^2\theta = 0$$

$$(1 - \sin^2\theta) - \sin\theta - \sin^2\theta = 0$$

$$1 - \sin\theta - 2\sin^2\theta = 0$$

$$2\sin^2\theta + \sin\theta - 1 = 0$$

$$(2\sin\theta - 1)(\sin\theta + 1) = 0$$

Either $2\sin\theta - 1 = 0$ or $\sin\theta + 1 = 0$

$\sin\theta = \frac{1}{2}$ $\sin\theta = -1$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ $\theta = \frac{3\pi}{2}$

We have $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$.

When $\theta = \frac{\pi}{6}$,

$V = 10\cos\frac{\pi}{6}\left(1 + \sin\frac{\pi}{6}\right)$

$= 10\left(\frac{\sqrt{3}}{2}\right)\left(1 + \frac{1}{2}\right)$

$= \frac{15\sqrt{3}}{2}\ \text{m}^3$

When $\theta = \frac{5\pi}{6}$,

$V = 10\cos\frac{5\pi}{6}\left(1 + \sin\frac{5\pi}{6}\right)$

$= 10\left(-\frac{\sqrt{3}}{2}\right)\left(1 + \frac{1}{2}\right)$

$= -\frac{15\sqrt{3}}{2}\ \text{m}^3$

When $\theta = \frac{3\pi}{2}$,

$V = 10\cos\frac{3\pi}{2}\left(1 + \sin\frac{3\pi}{2}\right)$

$= 10(0)(1 - 1)$

$= 0\ \text{m}^3$

The maximum possible volume of the trough is, $V = \frac{15\sqrt{3}}{2}$ m^3.

$\therefore$ The acute angle that would maximize the volume of the trough is $\theta = \frac{\pi}{6}$.

Total: 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.